

Article

Bi-Level Stackelberg Game-Based Optimization Model for Shared Energy Storage in Data Centers Considering Computational Flexibility

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Abstract

With the explosive growth in demand for artificial intelligence and computing power, the energy consumption of data centers has sharply increased, making their green and low-carbon operation an urgent need. Shared energy storage (SES), as a flexible regulation resource, can effectively facilitate renewable energy consumption and reduce system costs. However, existing research mostly regards data center loads as rigid loads, ignoring the elastic scheduling potential of latency-insensitive computing tasks, and lacking a game decision-making model from the perspective of SES operators to provide strategies for data center SES transactions. Therefore, this study constructs a SES trading optimization model that takes into account the flexibility of computing power. The upper-level targets profit maximization for the SES operator by optimizing charge and discharge strategies and service pricing, while the lower level minimizes the total energy cost of each computing center by jointly optimizing computing task scheduling and energy storage utilization plans. Finally, a bi-level Stackelberg game-based optimization model is constructed to solve the SES dispatch and, achieve benefit coordination between SES and computing center. The simulation results indicate that using SES without implementing load shifting cannot optimize the total cost of the computing power center, offering only storage revenue. Only by integrating SES with load shifting can a significant cost reduction be realized. The model presented in this article resulted in an SES revenue of 2605.34 yuan and a 3.14% reduction in computing center costs. This study provides a theoretical basis and decision-making support for shared energy storage participation in the computing power market and contributes to accelerating the coordinated development of computing power and electricity systems.



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1. Introduction

With the rapid advancement of artificial intelligence (AI) technology, the computing power industry has undergone explosive growth, giving rise to non-negligible energy

consumption challenges. Taking data centers as an example, national electricity consumption by data centers exceeded 150 billion kWh in 2023, accounting for 1.6% of total social electricity consumption, with the growth rate of energy demand becoming increasingly prominent. Under the explosive growth of computing power demand, the annual electricity consumption of data centers across the country is rapidly increasing. The traditional power supply mode that relies on long-distance transmission from the large power grid and significant peak–valley fluctuations not only fails to meet the rigid constraints of intelligent computing clusters on uninterrupted and highly reliable power, but also increases carbon emission intensity due to the high proportion of thermal power. In this context, it is urgently necessary to explore novel models including proximity renewable energy supply, aggregated trading, and on-site consumption, so as to increase the proportion of green electricity in data center operations, promote efficient green power-driven computing loads, and achieve safe, reliable, low-carbon, and economical operation objectives, thereby supporting high-quality development of data centers.

As a flexible regulation resource, shared energy storage demonstrates significant value in promoting renewable energy accommodation and reducing overall system costs. On the one hand, energy storage systems (ESS) can charge during off-peak electricity price periods and discharge during peak periods, effectively mitigating the peak–valley load difference and alleviating grid congestion. This arbitrage capability, coupled with the provision of ancillary services such as frequency regulation, enhances the economic viability of storage assets. On the other hand, data centers, as high-energy-consuming power users, can leverage energy storage to increase their proportion of green electricity access, thereby accelerating their green and low-carbon transition. By integrating ESS with power purchase agreements for renewable energy, data centers can shift flexible loads to periods of abundant renewable generation, enabling higher utilization of green electricity. Furthermore, shared energy storage enables multiple data centers within a cluster to collectively benefit from economies of scale, reducing individual investment costs and improving overall resource utilization efficiency. Preliminary studies have been conducted on the collaborative optimization of data centers and shared energy storage. Jiang et al. proposed a two-stage, bi-level collaborative framework to optimize the capacity planning of energy storage systems for computing centers under the direct green power supply mode, considering both investment and operational decisions [1]. Shared energy storage dispatch optimization models for data center clusters demonstrate the benefits of coordinated scheduling among multiple facilities [2,3]. These works largely concentrate on dispatch strategies and capacity configuration under deterministic or stochastic frameworks. However, the interplay between computing task scheduling flexibility and pricing mechanisms, particularly within a game-theoretic context, has not been fully addressed, leaving room for further investigation into the strategic interactions between shared energy storage operators and data center users in electricity markets.

Data centers can not only actively participate in demand response from the power side through energy storage systems, but also possess regulatory potential from the computing side. This computing-side flexibility arises from the temporal shiftability of specific workloads, which can be rescheduled to periods of lower electricity prices or higher renewable energy availability, complementing the physical flexibility provided by storage. Data center workloads can typically be classified into two categories: latency-sensitive tasks and latency-tolerant tasks. Latency-sensitive tasks, such as model training, offline inference, and batch processing jobs, only need to be completed before their deadlines, with processing time that can be flexibly scheduled within a certain window, exhibiting significant temporal flexibility [4]. This elasticity allows data centers to shift electricity consumption across time without violating service level agreements, creating new op-

opportunities for demand-side participation in electricity markets. Zhou et al. proposed a hierarchical load-modeling framework that simultaneously characterizes the energy consumption features of real-time interactive loads and latency-tolerant batch processing jobs, effectively achieving demand response while ensuring quality of service [5]. Zhou et al. further quantified the diverse real-time adjustable resources within both computing nodes and power nodes, and proposed a bi-level two-stage computing-electricity coordinated optimization dispatch model that unifies computing elasticity and energy storage resources within a single optimization framework, achieving efficient and coordinated utilization of computing-electricity resources [6]. While these works establish a solid foundation for computing-power coordination, they predominantly assume dedicated storage assets and do not investigate the strategic pricing interactions between a shared energy storage operator and multiple data center users. This gap highlights the need for a game-theoretic model that integrates computing flexibility into shared energy storage trading, which is the focus of this work.

A mathematical framework for analyzing strategic interactions among multiple decision-makers, game theory offers a systematic approach to multi-agent optimization problems characterized by conflicting or partially aligned interests. Unlike conventional centralized optimization methods, game theory does not require the assumption of complete information sharing or perfectly aligned objectives among participants. Instead, by explicitly modeling the decision variables, strategy spaces, and payoff functions of each agent, it seeks equilibrium solutions in which no participant has an incentive to unilaterally deviate from their chosen strategy—thereby more realistically capturing the decentralized, self-interested decision-making behavior observed in real-world systems. Among the various game-theoretic frameworks, the Stackelberg leader–follower game has attracted particular attention for its ability to capture the inherent hierarchical relationship that exists among decision-makers, in which the leader formulates its strategy by anticipating the follower’s optimal response, while the follower subsequently reacts optimally upon observing the leader’s decision, together forming a bi-level optimization problem. This modeling paradigm has been widely applied across domains such as resource allocation, dynamic pricing, and task scheduling to address multi-agent coordination problems characterized by information asymmetry and unequal decision-making power, offering a theoretical foundation for designing operational mechanisms that simultaneously safeguard the economic interests of individual participants and improve overall system efficiency.

Research in the energy market typically involves numerous stakeholders, such as power grids, consumers, and energy storage operators. Their interactions in power generation, transmission, trading, and consumption are often both competitive and cooperative, which is naturally suitable for game theory methods. In recent years, game theory has been widely applied to model typical interactions in electricity markets, for instance, pricing games between electricity retailers and consumers over time-of-use tariffs, and non-cooperative games among generators bidding for grid access. In shared energy storage transactions among data centers, the energy storage operator acts as the price setter, while individual data centers respond by scheduling their flexible loads and storage usage, naturally forming a Stackelberg game. The absence of such a pricing interaction model means that the mechanism for balancing storage revenue against data centers’ energy costs remains underexplored, thereby limiting the practical applicability of shared storage solutions in emerging computing power markets.

Therefore, this paper takes shared energy storage as the research object and fully considers the elastic scheduling characteristics of latency-insensitive batch processing tasks within data centers. We constructed a Stackelberg game optimization model for shared energy storage trading that explicitly accounts for computing flexibility. The upper level

aims to maximize the profit of the shared energy storage operator by jointly optimizing the charging and discharging strategies of the energy storage system and the service pricing offered to users. The lower level aims to minimize the comprehensive energy consumption cost of each computing center by collaboratively optimizing computing task scheduling and the plan for utilizing the leased storage capacity, subject to task deadlines and quality-of-service requirements. The interaction between the two levels captures the sequential decision process in which the operator first announces pricing, and then users react, with the operator anticipating their responses. The model is solved through a bi-level alternating iterative algorithm that converges to a Stackelberg equilibrium, achieving benefit coordination between the energy storage operator and computing center users. This work provides a theoretical basis and decision support for shared energy storage participation in the computing market, facilitating the efficient integration of computing flexibility and storage sharing toward cost-effective and low-carbon data center operations.

2. Literature Review

Compared with centralized energy storage, shared energy storage demonstrates significant advantages in utilization rate, cost sharing, and dispatch flexibility. Li et al. conducted an empirical comparison in residential scenarios and found that the annual utilization rate of energy storage devices can be improved under the shared model, while the energy cost per user is reduced, providing direct economic evidence for the promotion of shared energy storage [7]. Miller and Cariveau proposed that the synergy between shared energy storage dispatch strategies and market mechanisms is a key factor affecting the development of energy storage [8]. Xie et al. investigated the potential advantages of the multi-agent distributed shared energy storage service pattern in distribution networks [9]. Xiao et al. proposed an online coordinated dispatch framework for shared energy storage in interconnected data center microgrid scenarios, achieving energy complementarity among multiple microgrids [9]. For the business model, Kang et al. further proposed a two-stage multi-load collaborative shared energy storage model, realizing demand response optimization for multiple types of loads in data center clusters [10].

In the study of trading pricing mechanisms, game theory is the dominant approach. Jiang et al. adopted a cooperative game theory perspective and utilized the Shapley value for revenue allocation, proving the applicability of cooperative game theory in multi-microgrid shared energy storage scenarios [11]. Xu et al. proposed a bi-level optimization model for the shared hybrid hydrogen energy storage system to optimize the capacity configuration decisions and the pricing strategy jointly [12]. Li et al. proposed a stochastic optimized configuration method of SES in renewable energy power plants considering multiple primary–secondary game-pricing modes based on the uncertainty of renewable energy output [13]. However, most of the aforementioned trading models assume user loads as rigid demands and do not fully consider the dispatch potential of adjustable resources on the user side.

As high-energy-consumption entities, data centers exert a significant influence on the dispatch strategies of shared energy storage through their load characteristics. Although conventional studies typically treat data centers as rigid electricity consumers, recent research has begun to systematically quantify their flexibility resources. In recent years, research on computing-energy synergy has progressively deepened. Su et al. leveraged IoT technology to quantify the adjustable potential of data centers, providing data support for elastic computing scheduling [14]. Lu et al. proposed an online elastic resource provisioning framework with QoS guarantees, and theoretically proved that computing resources in container-based cloud scenarios can be elastically scheduled online while satisfying quality-of-service constraints [15]. Zhang et al. further systematically unlocked

data center flexibility by proposing a joint optimization method for energy storage dispatch and design under progressive loading, quantifying the regulation capacity that data centers can provide to the grid [16].

From the perspective of computing-energy collaborative optimization, Liu et al. jointly modeled the spatiotemporal migration characteristics of data center computing loads with shared energy storage dispatch, achieving computing-energy synergy optimization from a virtual power plant perspective [17]. Wu investigated the configuration and operation of data center shared energy storage from a stochastic programming perspective, considering the dual uncertainties of renewable generation and computing demand [18]. However, most existing studies on computing flexibility focus on internal optimization of individual data centers. Few works have incorporated computing elasticity into the shared energy storage trading framework, leaving the potential value of “computing + storage” synergy insufficiently explored.

The Stackelberg game is capable of characterizing the hierarchical decision-making relationship between leaders and followers in the market, and has been widely applied in energy-trading modeling. Its core advantage lies in the rational allocation of decision-making authority: the upper-level leader first sets prices or strategies, while the lower-level followers optimize their own decisions under the given strategies, with the equilibrium ultimately solved through backward induction. In the field of shared energy storage, Wang et al. applied a Stackelberg game to optimize the trading strategy of shared energy storage, proving that this method has improved economic benefits for all participants [19]. Jiang et al. adopted a cooperative game theory perspective and utilized the Shapley value for revenue allocation, which can serve as a methodological comparison to the bi-level game [11]. Lin et al. established a collaborative optimization dispatch model for multiple virtual power plants and shared energy storage considering electricity price uncertainty, employing robust optimization to handle price fluctuations; however, service quality was not addressed [20]. The aforementioned studies have validated the effectiveness of the bi-level game in energy storage trading across different scenarios, yet they share a common limitation: none of them consider the impact of user-side load flexibility on trading equilibrium. In data center scenarios, computing elasticity itself constitutes a tradable flexibility resource. Excluding it from the game framework not only underestimates the system’s regulation potential but also causes pricing strategies to deviate from the true equilibrium.

In addition, recent advances in intelligent optimization and machine learning methods, such as reinforcement learning based dynamic resource allocation [21], intelligent optimization of energy systems under varying operating conditions [22], and intelligent digital architectures and optimization strategies for complex industrial systems [23], which have demonstrated significant advantages in handling high-dimensional, nonlinear, and time-varying decision-making problems in energy management and resource scheduling. These studies provide cross-domain methodological support for the modeling approach adopted in this work, which embeds computational flexibility into the game equilibrium. At the same time, they also indicate that existing game-theoretic frameworks based on deterministic mathematical programming still have room for improvement when addressing dynamic, high-dimensional scenarios, an aspect identified as a direction for future research in this work.

From a comprehensive perspective, existing studies have made certain progress in the collaborative optimization of data centers and shared energy storage; however, the following limitations remain, as shown in Table 1. On one hand, the majority of studies treat data center loads as rigid loads, failing to fully exploit the elastic scheduling potential of latency-insensitive computing tasks. This oversight means that the temporal flexibility inherent in tasks such as model training, offline inference, and batch processing is not

harnessed to shift electricity consumption to periods of lower prices or higher renewable availability. Consequently, computing-side flexibility resources remain excluded from a unified optimization framework, leading to suboptimal coordination between power demand and storage dispatch [24,25]. On the other hand, existing works rarely construct decision models from the perspective of shared energy storage operators based on the game-theoretic relationship between operators and computing center users [26].

Table 1. Summary and comparison of existing literature.

Category	Approach	Advantages	Ref.	Limitations
Shared energy storage operation and trading mechanism	Empirical/operation-mode analysis	Provides economic evidence for shared ESS; identifies key research priorities	[9]	Does not address dynamic pricing-load interaction, and load-side adjustable resources not modeled
	Trading model/two-stage business model	Offers market-oriented operation reference; enables multi-load demand response	[10,16,20,27]	
Pricing mechanism	Cooperative game (Shapley)/Nash bargaining	Achieves fair revenue allocation and Pareto optimality	[11]	Load flexibility not integrated; lack of dynamic pricing mechanism for energy storage and computing center transactions
	Bi-level/non-cooperative game pricing	Validates bidding/pricing mechanisms in improving economics	[12,13]	
Data center operation	Computational flexibility; shiftable data load modeling	Quantifies data center flexibility potential	[14–18]	Lacks game-theoretic pricing interaction with ESS operator
	considering load flexibility and joint scheduling optimization with energy storage	Achieves computing-energy coordinated dispatch	[17]	
Stackelberg game applications	Stackelberg game/robust optimization	Validates bi-level game framework in multi-agent ESS trading	[13,19,20]	Excludes computing elasticity from game equilibrium

Therefore, this paper proposes a dual-layer game trading model for shared energy storage that takes into account computational flexibility. The contribution of the proposed model in this study is to incorporate the elastic scheduling potential of latency-insensitive computing tasks into the shared energy storage trading framework. Specifically, a Stackelberg game model is constructed, where the upper-level objective is to maximize the profit of the shared energy storage operator and the lower-level objective is to minimize the comprehensive energy cost of the data center. In this model, the upper level jointly optimizes the energy storage charging and discharging strategy and service pricing, while the lower level collaboratively optimizes the computing task scheduling and energy storage utilization plan. Case studies demonstrate that the proposed model can reduce the energy cost of data

centers while simultaneously improving the revenue of the shared energy storage operator, thereby validating the economic value of “computing + storage” synergistic optimization.

3. Problem Formulation

The operation framework of the data center and shared energy storage is illustrated in Figure 1. As shown in the figure, the grid-distributed photovoltaic (PV) systems, distributed wind power, and energy storage station modules jointly supply electricity to the data center. The energy storage system can serve as a flexible resource in the electricity market by purchasing low-cost electricity during peak periods for storage and subsequently discharging it during demand peak periods of the computing center, thereby balancing supply and demand, reducing energy waste, and capturing revenue. In this paper, the energy storage system is regarded as an independent market participant, equivalent to other market entities, providing services to multiple data centers within the park under a shared mode.

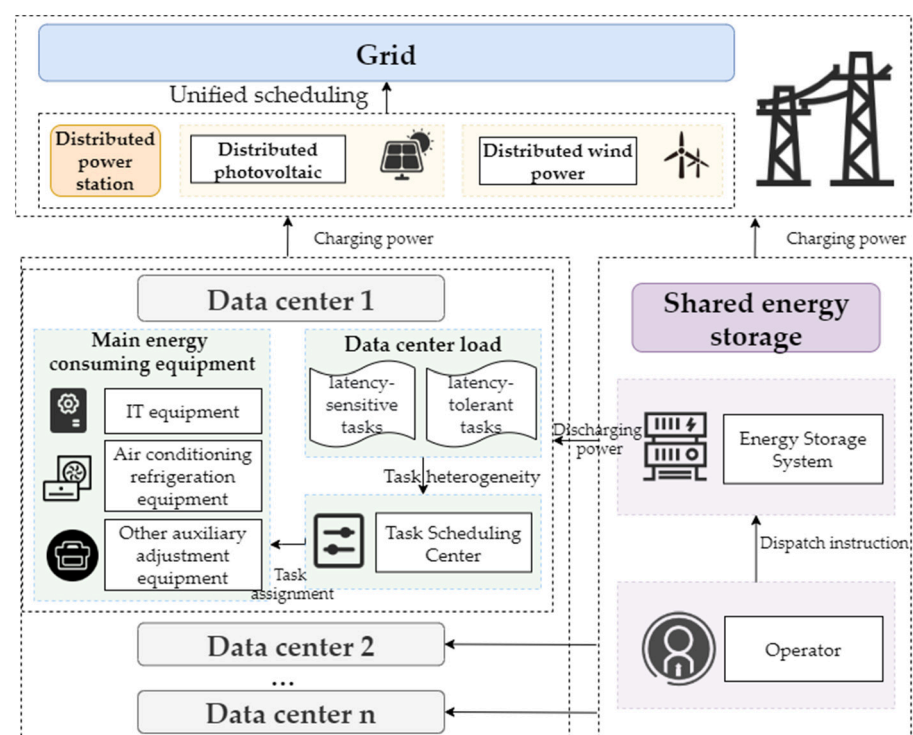


Figure 1. Data center operation framework diagram.

Although the framework above outlines the basic operational logic of the system, the following key challenges remain in the coordinated operation of shared energy storage and multiple data centers:

- (1) Interest coordination and gaming challenge. The energy storage operator aims to maximize arbitrage revenue, while each data center aims to minimize its electricity procurement cost. These objectives are conflicting yet interdependent, making it necessary to establish an effective coordination mechanism.
- (2) Heterogeneous load-scheduling challenge. Data centers contain both latency-sensitive tasks and shiftable data loads (SDL). Without a unified pricing signal, coordinating load scheduling across multiple data centers to achieve system-wide peak-shaving and valley-filling remains a systematic challenge.
- (3) Absence of a dynamic pricing mechanism. Existing studies often treat the storage operator’s pricing decisions and the data centers’ scheduling responses separately,

lacking a bi-level dynamic mechanism that captures their strategic interaction, making it difficult to characterize the system's equilibrium under multi-agent gaming.

To address the above challenges, this paper aims to:

- Establish a Stackelberg game-based bi-level decision-making framework, in which the energy storage operator acts as the leader setting the internal transaction price, and the data centers act as followers optimizing their load-scheduling strategies, thereby addressing the interest coordination challenge.
- Develop a shiftable load-scheduling optimization model for data centers that distinguishes between latency-sensitive and latency-tolerant tasks, enabling coordinated load scheduling driven by the pricing signal, thereby addressing the heterogeneous load-scheduling challenge.
- Derive and analyze the Stackelberg equilibrium of the proposed game, demonstrating that the equilibrium pricing and scheduling strategies can jointly optimize storage revenue and data center costs, thereby addressing the absence of a dynamic pricing mechanism.

4. Methodology

Based on the aforementioned problem description and operational framework, this paper constructs a bi-level decision-making model for the collaborative optimization between the shared energy storage (SES) operator and data centers, aiming to achieve coordinated optimal dispatch of shared energy storage and computing resources. In this bi-level decision-making model, the SES operator serves as the upper-level leader, while the data centers act as lower-level followers.

4.1. Upper-Level Decision Model

The upper-level decision model maximizes the profit of the shared energy storage (SES) operator by transmitting electricity price signals and demand response instructions to the lower-level data centers, thereby incentivizing the data centers to adjust their computing loads across different time periods and guiding them to optimize their own electricity consumption decisions.

4.1.1. Objective Function

The objective function of the upper-level decision model comprises three components, namely the total revenue from electricity sales by the shared energy storage (SES) operator $I_{t,s}^{dis}$, the electricity procurement cost from the grid $C_{t,s}^{ch}$, and the charging/discharging loss cost $C_{t,s}^{lo}$. Equation (1) is used to formulate the overall objective function. Since fixed investment costs and fixed loss costs remain constant regardless of decision variable changes and do not affect the optimal solution, these fixed costs are excluded from the model design. Equation (3) is used to calculate the electricity purchasing cost $C_{t,s}^{lo}$, which depends on the procurement volume of the shared energy storage operator and the current market electricity price [28]. The charging/discharging loss cost of the energy storage system refers to the cost of energy losses incurred during the charging and discharging processes, where the energy conversion efficiency cannot reach 100% [29]. Equation (4) is used to compute the charge and discharge loss cost.

$$\max R = I_t^{dis} - C_t^{ch} - C_t^{lo} \quad (1)$$

$$I_t^{dis} = P_t^{dis} \cdot \lambda_t^{dis} \quad (2)$$

$$C_t^{ch} = P_t^{ch} \cdot \lambda_t^{ch} \quad (3)$$

$$C_t^{lo} = \lambda_t^{ch} P_t^{ch} (1 - \eta^{ch}) \Delta T + \lambda_t^{dis} P_t^{dis} \left(\frac{1}{\eta^{dis}} - 1 \right) \Delta T \quad (4)$$

where $P_{t,s}^{ch}$ denotes the charging power of the energy storage system at time t under scenario s ; η^{ch} represents the charging efficiency; η^{dis} represents the discharging efficiency; and ΔT denotes the time interval.

4.1.2. Constraints

(1) Charge and discharge power constraint

The charging and discharging power of the energy storage system must not exceed the rated power limits. Equations (5)–(8) are used to formulate the corresponding power constraints of the energy storage system.

$$P_t^{ch,e} = P_t^{ch} \cdot \eta_{ch} \quad (5)$$

$$P_t^{dis} = P_t^{dis,e} \cdot \eta_{dis} \quad (6)$$

$$\begin{cases} P_t^{dis} = \sum_{i=1}^I P_{i,t}^{dis} \\ P_t^{ch} = \sum_{i=1}^I P_{i,t}^{ch} \end{cases} \quad (7)$$

$$\begin{cases} 0 \leq P_t^{dis} \leq P^{\max} \cdot \mu_t^{dis} \\ 0 \leq P_t^{ch} \leq P^{\max} \cdot \mu_t^{ch} \end{cases} \quad (8)$$

$$\mu_t^{ch} + \mu_t^{dis} \leq 1, \forall t \quad (9)$$

where η_{ch} and η_{dis} represent the charging and discharging efficiencies of the battery, respectively; $P_{i,t}^{dis}$ and $P_{i,t}^{ch}$ represent the electricity sold to and purchased from data center i , respectively; P_t^{dis} and P_t^{ch} denote the total electricity sold and purchased by the SES operator during time period t , respectively; $P_t^{dis,e}$ and $P_t^{ch,e}$ represent the actual discharging and charging power of the ESS during time period t , respectively; μ_t^{ch} and μ_t^{dis} are binary variables, where $\mu_t^{ch} = 1$ indicates that the energy storage system is charging and $\mu_t^{dis} = 1$ indicates that it is discharging; T is the total number of time periods, set to 24 in this paper; and I denotes the number of data centers within the service scope of the shared energy storage. Since the charging and discharging operations of the energy storage system cannot be performed simultaneously, Equation (9) is used to enforce the mutual exclusivity constraint, where μ_t^{ch} and μ_t^{dis} are binary variables representing the charging and discharging states of the energy storage system at time t , respectively.

(2) Energy storage state of charge (SOC) constraints

The state of charge (SOC) of the energy storage battery is calculated using Equations (10) and (11). The SOC is required to remain within the prescribed minimum and maximum limits. Equation (12) is used to formulate the SOC constraint.

$$SOC_t = \frac{S_t}{S_{\max}} \quad (10)$$

$$S_t = S_{t-1} \cdot (1 - \phi) - P_t^{dis,e} \cdot \Delta t + P_t^{ch,e} \cdot \Delta t \quad (11)$$

$$SOC_{\min} \leq SOC_t \leq SOC_{\max} \quad (12)$$

where S_t denotes the SOC of ESS during time period t ; ϕ is the self-discharge coefficient of the energy storage; $SOC_{\min}$ and $SOC_{\max}$ represent the maximum and minimum SOC limits, respectively.

In addition, to ensure the sustainability of the dispatching schedule, the total charging and discharging energy must be balanced over the entire scheduling cycle. Equation (13) is used to enforce the energy balance constraint.

$$SOC_1 \cdot (1 - \varepsilon) \leq SOC_{24} \leq SOC_1 \cdot (1 + \varepsilon, s) \quad (13)$$

where SOC_1 is the initial SOC of the ESS; SOC_{24} is the SOC at the end of the intra-day scheduling period; ε is the tolerance of the final charging state, meaning that the final SOC at the end of the day should remain within the tolerance range of the initial SOC.

(3) Internal pricing constraints

The shared energy storage operator adopts an internal electricity pricing mechanism to facilitate electricity trading with data centers. The internal price is set with reference to the grid electricity price [30], and must satisfy the following constraint.

$$\lambda_t^f \leq \lambda_t^{dis} \leq \lambda_t \quad (14)$$

where λ_t^f denotes the feed-in tariff at time period t ; λ_t^{dis} represents the internal selling price set for data centers at time period t , respectively; and λ_t is the grid electricity price charged to data centers.

4.2. Lower-Level Decision Model

The lower level constitutes the data center decision optimization model. The objective of the lower-level model is to minimize the total operating cost of the data centers. Each data center achieves cost optimization by adjusting its computing task allocation strategy.

4.2.1. Objective Function

The objective function comprises the electricity procurement cost from the grid and the electricity procurement cost from the shared energy storage (SES). The detailed formulation is in Equations (15)–(17).

$$\min C_{IDC} = C_{grid} + C_{ess} \quad (15)$$

$$C_{grid} = \sum_{t=1}^T \sum_{i=1}^I P_{i,t}^G \cdot \gamma_t \quad (16)$$

$$C_{ess} = \sum_{t=1}^T \sum_{i=1}^I P_{i,t}^{dis} \cdot \gamma_t^{dis} \quad (17)$$

where $P_{i,t}^G$ denotes the electricity procurement power of data center i from the grid at time t ; $P_{i,t}^{dis}$ denotes the electricity procurement power of data center i from the SES at time t .

4.2.2. Constraints

(1) Load-processing capacity constraint

The workload processed by each data center at each time period is limited by the number of servers. The total workload must not exceed the prescribed limit. Equation (18) is used to formulate the upper-bound constraint. In addition, the dispatched load at each time period is required to be non-negative and must not exceed its upper bound. Equations (19) and (20) are used to enforce the non-negativity and upper-bound constraints, respectively. After the data load processing is completed, it is fed back to the user in the form of a data stream. Due to the limitation of network bandwidth during the transmission

process, task delays are inevitable. To ensure service quality, constraints are set as shown in Equation (21)

$$0 \leq D_{i,t} \leq D_{i,\max}^{SER} \quad (18)$$

$$\sum_{t''=t+1}^T D_{i,t \rightarrow t''}^{Tran} \leq D_{i,t}^{Tran_0} \quad (19)$$

$$0 \leq D_{i,t \rightarrow t''}^{Tran} \quad (20)$$

$$\frac{D_{i,t}}{N_i^{SER} \cdot B} \leq \Delta t \quad (21)$$

where $D_{i,\max}^{SER}$ denotes the maximum workload-processing capacity of data center i at time period t . $D_{i,t \rightarrow t''}^{Tran}$ denotes the workload shifted from time period t to time period t'' . $D_{i,t}^{Tran_0}$ denotes the maximum workload that data center i can shift in each time period. N_i^{SER} denotes the number of servers in each data center. B denotes the bandwidth (kB/s) of a single server connected to the user's network.

(2) Power balance constraints

In practical applications, the model must also satisfy power balance constraints. These constraints couple the data center's electricity consumption power, the energy storage system power, and the grid supply power, ensuring that electricity demand and supply remain balanced in each time period. The specific constraints are formulated in Equations (22)–(25).

$$P_{i,t}^{IDC} + P_{i,t}^{dis} = P_{i,t}^{PV} + P_{i,t}^{WT} + P_{i,t}^G - P_{i,t}^{AB} \quad (22)$$

$$P_{i,t}^G \geq 0 \quad (23)$$

$$P_{i,t}^{dis} \geq 0 \quad (24)$$

$$P_{i,t}^{AB} \geq 0 \quad (25)$$

where $P_{i,t}^{PV}$ and $P_{i,t}^{WT}$ denote the distributed photovoltaic and wind power generation at time t , respectively; $P_{i,t}^G$ represents the power exchanged between data center i and the grid, which is constrained to be non-negative to ensure unidirectional power purchase from the grid. $P_{i,t}^{AB}$ represents the amount of abandoned photovoltaic and wind power.

4.3. Computational Flexibility Modeling

The energy consumption of data centers primarily originates from IT equipment, air conditioning (cooling) equipment, and other auxiliary regulatory devices [31]. Given the minor energy contribution of the auxiliary devices, the data center's overall power usage can be modeled as a sum of individual components, as formalized by Equations (26)–(28).

$$P_{i,t}^{IDC} = \frac{P_{i,t}^{SER} + P_{i,t}^{ACS}}{\eta_i^{IDC}} \quad (26)$$

$$P_{i,t}^{SER} = \frac{k \cdot f^3}{\mu} \cdot D_{i,t} \quad (27)$$

$$P_{i,t}^{ACS} = \frac{\varepsilon}{\eta_{ACS}} \cdot P_{i,t}^{SER} \quad (28)$$

where Equation (27) represents the IT equipment energy consumption model. The IT equipment power is proportional to the cube of the central processing unit (CPU) operating frequency and to the workload to be processed by the data center in time period t . Equation (28) represents the air conditioning cooling equipment energy consumption model, which

is calculated based on the linear relationship between the IT equipment electric power and the cooling equipment cooling power. $P_{i,t}^{SER}$ denotes the IT equipment energy consumption; $P_{i,t}^{ACS}$ denotes the air conditioning cooling equipment energy consumption; η_i^{IDC} represents the data center energy ratio, defined as the proportion of the sum of IT and cooling energy consumption to the total energy consumption; k is the dynamic energy consumption coefficient; f is the CPU operating frequency (GHz); μ is the average service rate of the servers (%); $D_{i,t}$ denotes the final total workload in time period t (GB); ε is the conversion coefficient, specifically the ratio of the cooling equipment cooling power to the IT equipment electric power; η^{ACS} is the energy efficiency coefficient of the air conditioning equipment, defined as the ratio of cooling power to electric power.

The computational workload of a data center, defined as the processing load required in each time period, can be classified into shiftable data load (SDL) and rigid load according to the response time requirements of the demand response services provided to customers.

$$D_{i,t} = D_{i,t}^0 + \sum_{t'=1}^{t-1} D_{i,t' \rightarrow t}^{Tran} - \sum_{t''=t+1}^T D_{i,t \rightarrow t''}^{Tran} \quad (29)$$

where $D_{i,t}^0$ denotes the initial workload of data center i at time period t .

Shiftable data load (SDL), also referred to as batch-processing load, has no strict deadline constraint and only requires completion within a specified time window. Typical examples include large model pre-training, offline batch inference, and data preprocessing. The SDL is calculated in Equations (30) and (31).

$$D_{i,t}^{Tran0} = \beta^{Tran} \cdot D_{i,t}^0 \quad (30)$$

$$D_{i,t}^{Tran} = D_{i,t}^{Tran0} + \sum_{t'=1}^{t-1} D_{i,t' \rightarrow t}^{Tran} - \sum_{t''=t+1}^T D_{i,t \rightarrow t''}^{Tran} \quad (31)$$

where β^{Tran} denotes the curtailable load ratio.

Rigid load tasks have zero tolerance for latency and must be executed in real time at the designated moment. Typical examples include interactive inference requests and real-time video stream processing. The rigid load is calculated in Equation (32).

$$D_{i,t}^{Rigi} = (1 - \beta^{Tran}) \cdot D_{i,t}^0 \quad (32)$$

4.4. Solving Method

Since the lower-level objective function contains nonlinear terms, specifically the product of internal electricity prices and charging/discharging power, this paper adopts a numerical optimization method based on the Karush–Kuhn–Tucker (KKT) conditions [32] to convert the bi-level optimization problem into a mixed-integer linear programming (MILP) problem for solution. The decision variables include the continuous variables: the data center's power purchased from the grid $P_{i,t}^G$, the power purchased from the shared energy storage $P_{i,t}^{dis}$, the data center's final total workload $D_{i,t}$, and the transferable load at time period t , denoted $D_{i,t}^{Tran}$.

Let the dual variables of the linear programming constraints Equations (18)–(25) be denoted as $\mu_{i,t}^-$, $\mu_{i,t}^+$, $\beta_{i,t}$, $\pi_{i,t}$, $\theta_{i,t}$, $\sigma_{i,t}$, $v_{i,t}$, and $\delta_{i,t}$, respectively. The Lagrangian function of

the lower-level problem can then be constructed, yielding the following KKT conditions for the above linear programming formulation.

$$\begin{aligned}
 & \gamma_t + \theta_{i,t} - \sigma_{i,t} = 0, \forall t, \forall i \\
 & \gamma_t^{dis} - \theta_{i,t} - v_{i,t} = 0, \forall t, \forall i \\
 & \alpha_{i,t} \cdot (\theta_{i,t} - \theta_{i,t \rightarrow t''}) + (\mu_{i,t}^- + \mu_{i,t}^+) - (\mu_{i,t \rightarrow t''}^- + \mu_{i,t \rightarrow t''}^+) + \beta_{i,t} - \pi_{i,t} = 0 \\
 & 0 \leq \mu_{i,t}^- \perp D_{i,t} \geq 0, \forall t, \forall i \\
 & 0 \leq \mu_{i,t}^+ \perp (D_{i,\max}^{SER} - D_{i,t}) \geq 0, \forall t, \forall i \\
 & 0 \leq \beta_{i,t} \perp (D_{i,t}^{Tran0} - \sum_{t''=t+1}^T D_{i,t \rightarrow t''}^{Tran}) \geq 0, \forall t, \forall i \\
 & 0 \leq \pi_{i,t} \perp D_{i,t \rightarrow t''}^{Tran} \geq 0, \forall t, \forall i \\
 & P_{i,t}^{IDC} - P_{i,t}^{dis} - P_{i,t}^{PV} - P_{i,t}^{WT} - P_{i,t}^G + P_{i,t}^{AB} = 0, \forall t, \forall i \\
 & 0 \leq \sigma_{i,t} \perp P_{i,t}^G \geq 0, \forall t, \forall i \\
 & 0 \leq v_{i,t} \perp P_{i,t}^{dis} \geq 0, \forall t, \forall i \\
 & 0 \leq \delta_{i,t} \perp (\Delta t - \frac{D_{i,t}}{N_i^{SER} \cdot B}) \geq 0, \forall t, \forall i
 \end{aligned} \tag{33}$$

By introducing binary variables $z_1, z_2, \dots, z_7$ and applying the Big-M method, the nonlinear constraints are transformed into the following linear inequalities.

$$\begin{aligned}
 & 0 \leq \mu_{i,t}^- \leq M \cdot z_1 \\
 & 0 \leq D_{i,t} \leq M \cdot (1 - z_1) \\
 & 0 \leq \mu_{i,t}^+ \leq M \cdot z_2 \\
 & 0 \leq D_{i,\max}^{SER} - D_{i,t} \leq M \cdot (1 - z_2) \\
 & 0 \leq \beta_{i,t} \leq M \cdot z_3 \\
 & 0 \leq D_{i,t}^{Tran0} - \sum_{t''=t+1}^T D_{i,t \rightarrow t''}^{Tran} \leq M \cdot (1 - z_3) \\
 & 0 \leq \pi_{i,t} \leq M \cdot z_4 \\
 & 0 \leq D_{i,t \rightarrow t''}^{Tran} \leq M \cdot (1 - z_4) \\
 & 0 \leq \sigma_{i,t} \leq M \cdot z_5 \\
 & 0 \leq P_{i,t}^G \leq M \cdot (1 - z_5) \\
 & 0 \leq v_{i,t} \leq M \cdot z_6 \\
 & 0 \leq P_{i,t}^{dis} \leq M \cdot (1 - z_6) \\
 & 0 \leq \delta_{i,t} \leq M \cdot z_7 \\
 & 0 \leq \Delta t - \frac{D_{i,t}}{N_i^{SER} \cdot B} \leq M \cdot (1 - z_7)
 \end{aligned} \tag{34}$$

5. Case Study

5.1. Basic Data and Parameters

To validate the proposed model and algorithm, simulation experiments are conducted based on the scale of a real-world data center, with partial parameters adopted from [33,34]. Three data centers (IDC1, IDC2, IDC3), constructed and managed by the same operator, are considered. The three data centers are geographically adjacent and share identical hardware components and performance parameters across their servers. The scheduling time step is set to 1 h, with a unit-scheduling horizon of 1 day. Data center energy consumption parameters are referenced from [17]. The shared energy storage system constructed in this paper employs lithium-ion (Li-ion) batteries as the storage medium. The shared energy storage system has a total power capacity of 1000 kW and an energy capacity of 6000 kWh. The state of charge (SOC) is constrained within [0.1, 0.9], with an initial SOC of 0.5. The self-discharge coefficient is set to 0.04% per time period 28. Both charging and discharging

efficiencies are 0.95. The terminal SOC tolerance is set to 5% [28]. The model is solved using Python 3.8 with the Gurobi optimizer.

5.2. Result Analysis

Based on the data center parameters and SES configurations specified above, the optimization yields the SES charging and discharging trading strategy that accounts for computing flexibility. The results are illustrated in Figure 2. As shown in Figure 2, the SES exhibits two typical charge and discharge cycles. The first cycle occurs approximately during periods 3 to 11, during which the SES starts with a low-power charging at time 3, then rapidly charges at full power at time 4 until reaching the upper limit, and discharges to the data center during periods 7 and 11. The second cycle occurs approximately during periods 14 to 21, during which the ESS undergoes full-power charging again from periods 14 to 18, and discharges power during periods 19 and 21. It can be observed that the charging of the SES is concentrated during valley and flat electricity price periods, while discharging occurs during peak price periods. This enables the SES operator to capture arbitrage profits by purchasing electricity at low prices and selling it at high prices.

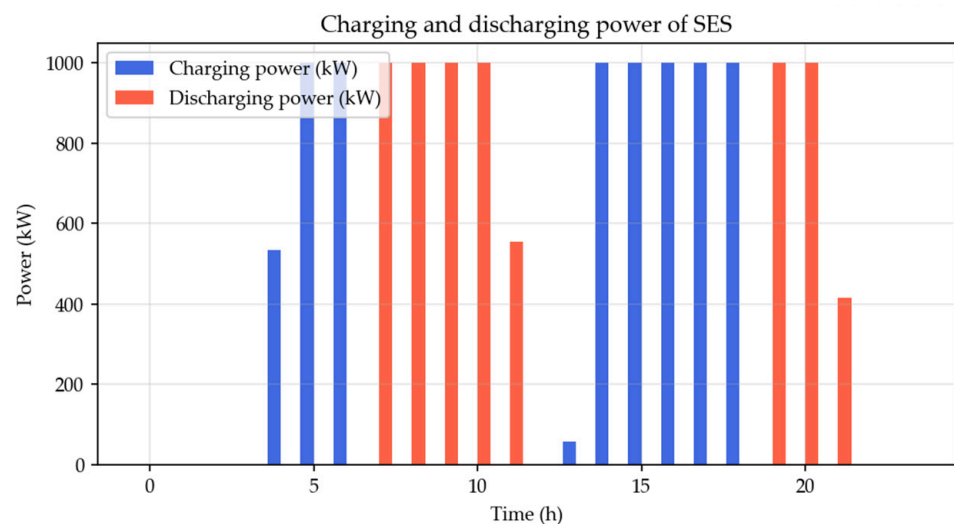


Figure 2. Charging and discharging power of SES.

Figure 3 illustrates the daily variation curve of SOC for ESS. The SES primarily executes a single valley-charging and peak-discharging operation. The SOC remains within the safe operating range of [0.1, 0.9] throughout the entire day. Specifically, the SOC maintains its initial level of approximately 0.5 during periods 0–3. During periods 4–7, the SOC rises rapidly to the upper limit of 0.9 under high-power charging. Subsequently, the SOC declines to approximately 0.1 due to continuous discharging. A second charging cycle is initiated in period 13, during which the SOC recovers to approximately 0.9. Thereafter, the SOC gradually decreases to approximately 0.48 and stabilizes during periods 20–23. Owing to the coordination of flexible computing tasks, the SES does not need to shoulder the entire peak-shaving burden alone. The depth of discharge cycles is moderately reduced, which effectively mitigates electrochemical degradation of the battery and contributes to extending its economic lifetime.

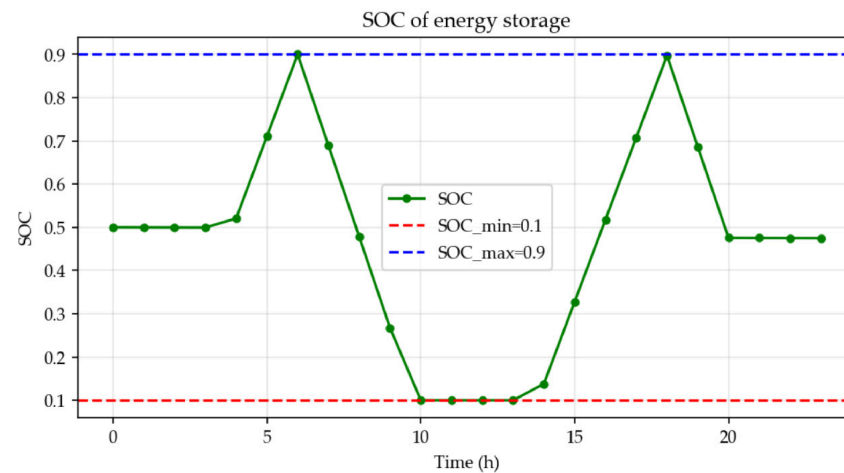


Figure 3. Daily variation curve of SOC for ESS.

Figure 4 illustrates the scheduling distribution of shiftable data load across all time slots throughout the day for the computing data centers. The SDL of DC1 remains at a consistently low level over the entire day, with only minor fluctuations observed during time slot 6 and slots 10–18. The SDL of DC3 exhibits minor fluctuations at time slot 1–6, followed by a second significant peak at time slot 12. The SDL of DC2 is concentrated in the 12–23 and 0–6 time periods, indicating that a substantial volume of latency-insensitive tasks is concentrated and deferred to the late-night and early morning off-peak periods for execution. The coordinated SDL scheduling across the three data centers collectively achieves temporal load shifting of the computing workload, thereby alleviating the grid power procurement pressure during peak periods.

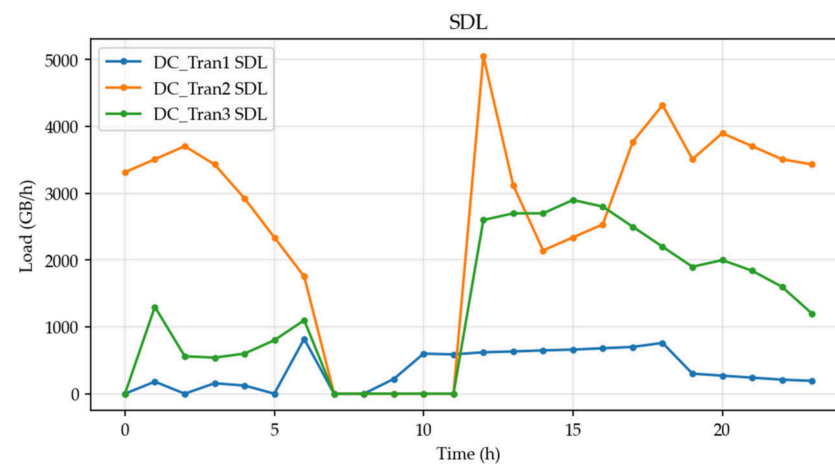


Figure 4. Shiftable data load variation curve of computing power center.

The above results demonstrate the performance of the SES and computing power centers. To further validate the effectiveness of the model, the following Discussion section proves it through multi-scenario comparison and sensitivity analysis.

5.3. Discussion

(1) Comparison of multiple scenes

To verify the effectiveness of the model, this paper designed two benchmark scenarios for comparison with the model presented in this paper. Scenario 1 is without the shared energy storage and also do not consider the flexibility of computing power during the operation of the data centers. Scenario 2 with shared energy storage services, but without

considers the flexibility of data center computing power. The model in this study is set to Scenario 3. Under unchanged parameters, the simulation results of the three scenarios are shown in Table 2.

Table 2. Comparison results of multiple scenarios.

Scenarios	Total Cost of DC (10^4 Yuan)	Energy Storage Revenue (Yuan)
Scenario 1	72.13	-
Scenario 2	72.13	2605.34
Scenario 3	69.93	2605.34

The proposed model, Scenario 3, leverages the synergy of energy storage and load shifting to achieve the lowest total cost of 69.93 ten thousand yuan along with an energy storage revenue of 2605.34 yuan. Scenario 1, without storage or load shifting, and Scenario 2, with storage but no load shifting, exhibit nearly identical total costs of approximately 72.13 ten thousand yuan, an increase of about 3.14 percent. Scenario 1 yields no energy storage revenue, whereas Scenario 2 obtains the same revenue as the proposed model due to its storage installation but fails to reduce the total cost. This demonstrates that merely adding energy storage without implementing load shifting cannot optimize the total cost of the computing power center, offering only storage revenue. Only by integrating energy storage with load shifting can a significant cost reduction be realized, highlighting the effectiveness of the proposed model.

(2) Sensitivity analysis

To further validate the robustness of the proposed model (Scenario 3) and to reveal the impact mechanisms of key parameters on DC operating cost and ESS operator revenue, this section conducts a sensitivity analysis on five key parameters—electricity price coefficient, ESS capacity coefficient, battery charging and discharging efficiency, load computing flexibility coefficient, and renewable energy output coefficient—under three scenarios. The results are presented in Figure 5.

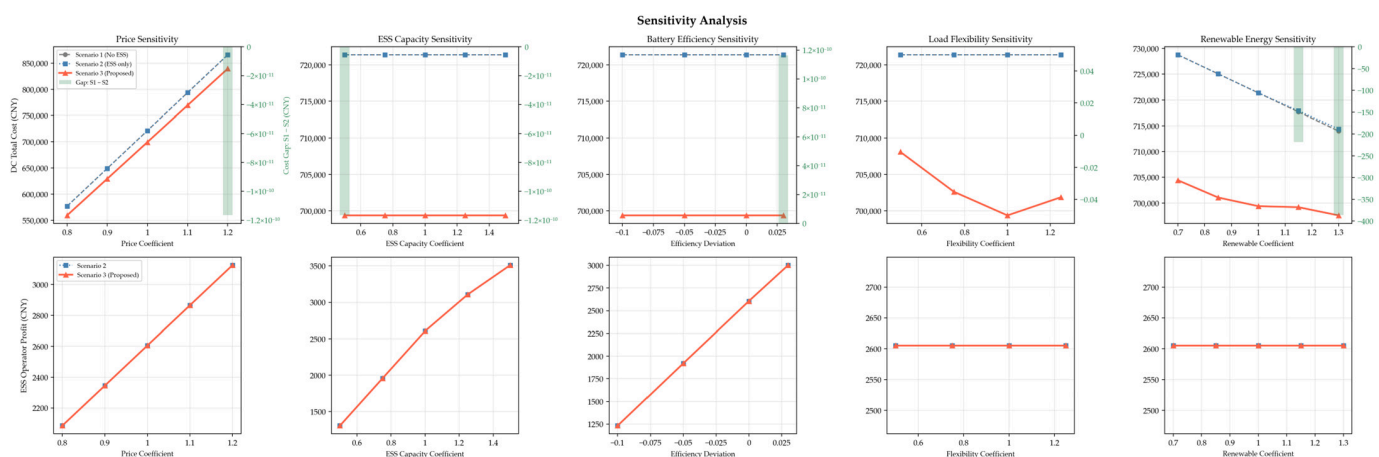


Figure 5. Sensitivity analysis results.

In summary, the sensitivity analysis results demonstrate that the proposed model (Scenario 3) consistently outperforms Scenario 1 and Scenario 2 in terms of cost control capability and ESS profitability under various uncertainty scenarios, including electricity price fluctuations, ESS capacity configuration, battery efficiency degradation, load flexibility variation, and renewable energy penetration changes. These findings confirm the effectiveness and robustness of the proposed model.

6. Conclusions

This paper addresses the shared energy storage (SES) trading optimization problem for data center clusters, aiming to exploit the temporal flexibility of latency-insensitive computing tasks. A bi-level Stackelberg game-based optimization model that accounts for computing workload flexibility is proposed. The principal findings are summarized below. First, a bi-level optimization model incorporating computing load flexibility is proposed. The upper-level model maximizes the SES operator's profit by optimizing the charging and discharging strategy and internal electricity pricing. The lower-level model minimizes the comprehensive energy cost of each data center by coordinating the scheduling of latency-insensitive computing tasks with the SES power procurement plan.

Second, the temporal scheduling of shiftable data load demonstrates a pronounced peak-shaving effect. By proactively migrating latency-insensitive tasks, such as model training and offline inference, during valley price periods, data centers effectively reduce rigid electricity demand during peak periods, thereby decreasing the volume of high-price electricity purchased directly from the grid and improving the overall energy cost efficiency of the system.

Third, the bi-level pricing mechanism achieves interest coordination between the SES operator and the data center users. Through internal electricity pricing, the SES offers discounted rates below the grid price during peak periods to incentivize data centers to purchase from the SES, while encouraging flexible charging during valley periods. This ensures the SES operator's arbitrage margin while providing data centers with economic incentives to participate in demand response, forming a benign market interaction mechanism.

In summary, the proposed computing-electricity coordinated bi-level optimization model effectively integrates the regulation capability of shared energy storage with the temporal flexibility of computing workloads, demonstrating significant advantages in reducing data center comprehensive energy costs, enhancing SES operational revenue, and promoting green energy consumption. Future work may further extend the current deterministic bi-level framework toward a stochastic or robust bi-level formulation, potentially informed by machine-learning-based approaches discussed above for handling high-dimensional uncertainty, including renewable generation forecasting errors. In addition, incorporating more detailed server-level operational constraints, such as server utilization limits, communication overhead and delays, task migration costs, and SLA violation penalties, would further refine the computing-side flexibility model. Battery degradation effects and more realistic, time-varying workload representations could also be incorporated to improve the fidelity of the proposed framework.

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